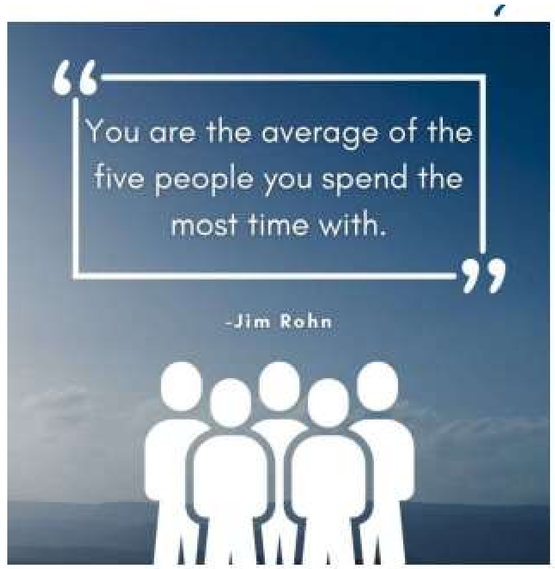
KNN Intuition

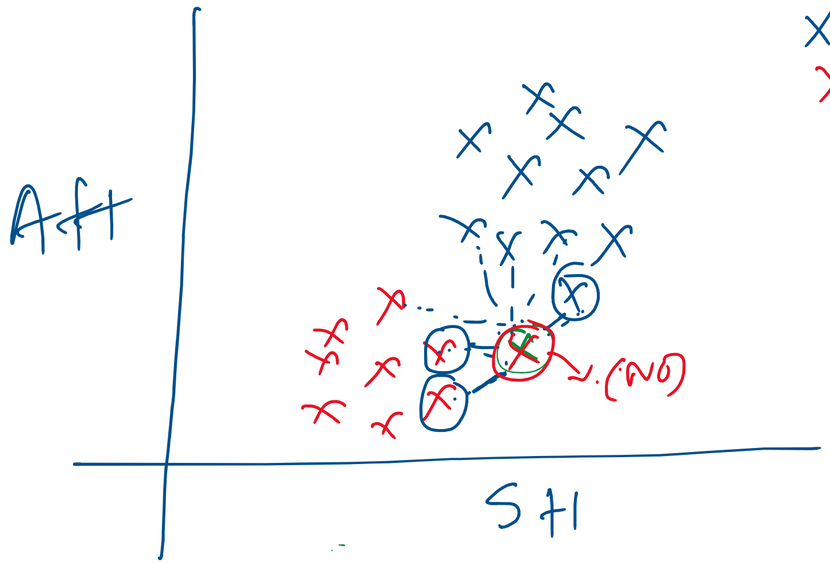


→ philosophy
↓ based
KNN

Study Hours (1)	Attendance (2)	Target Pass
8	95	Yes
7	90	Yes
2	50	No
1	45	No
5	70	Yes

SH | Att

New data → Model → predict → (Yes/No)



X → Yes
X → No

K = 3

500 data

Distance →
Sort →
majority Count →

X → Yes
X → No
X → NO

→ No

euclidean | Manhattan Distance

2D → 3D → 4D → 5D → .... ND

How select K

Heuristic

Experimental

$k \simeq \sqrt{n}$

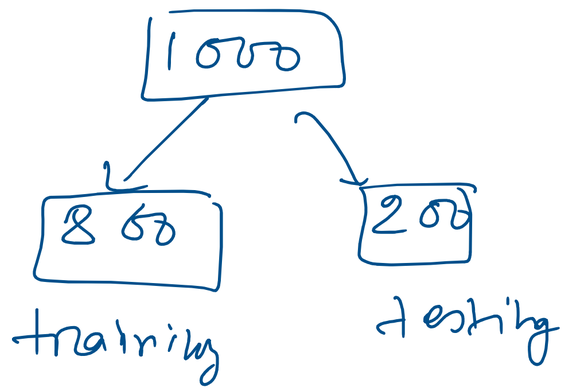
n = number of training sample

100

$K \simeq \sqrt{100} = 10$

1, 2, 3, 4, .. 10

Experimental



→ KNN = 1 → Predict → 85

KNN = 2 → 90

KNN = 3 → 93 → K = 3

KNN = 4 → 75

. . . .

. . .

KNN = 150 → 60

K = 1 → KNN → Accuracy → 80

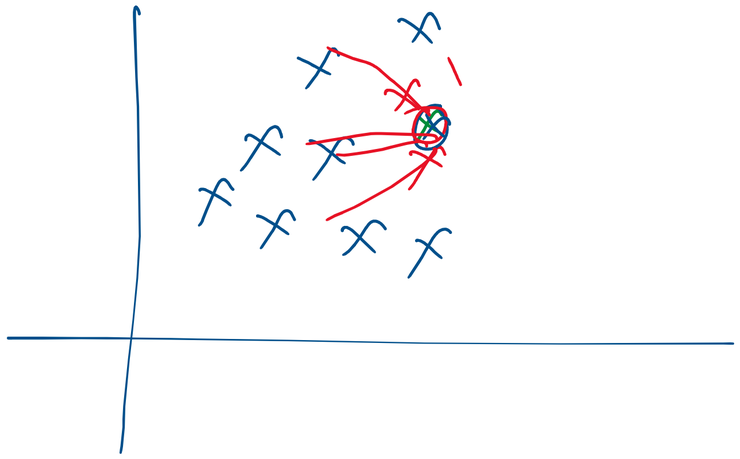
K = 2 . . — — 70

K = 3

K = 3

training

= 10



K = 10

100

K = 100

Sort

1 → Yes
2 → Yes
3 → Yes
4 → No
5 → No
6 → Yes
7 → Yes
8 → Yes
9 → Yes
10 → Yes

→ Yes

Limitation of KNN

1) Largest Dataset, n = 10L, f = 150

Lazy Learning Technique

training = Nothing = Fast

Training = Nothing = fast

→ prediction = Done all computation → slow

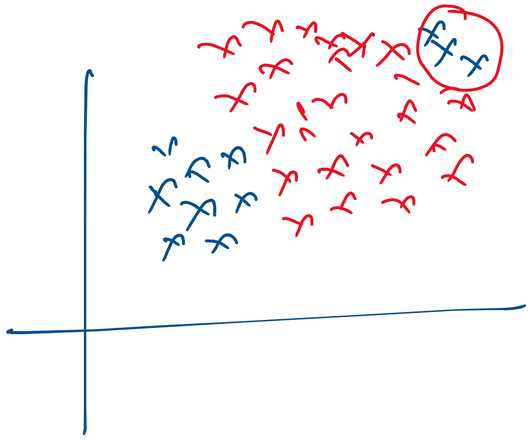
2) High Dimensional data → f = 600

(curse of dimensional) → Distance len meaningful

→ all points look equally far

KNN → Distance → Prediction

3) Outliers → sensitive



4) Sensitive to feature scaling

Age (0, 100)

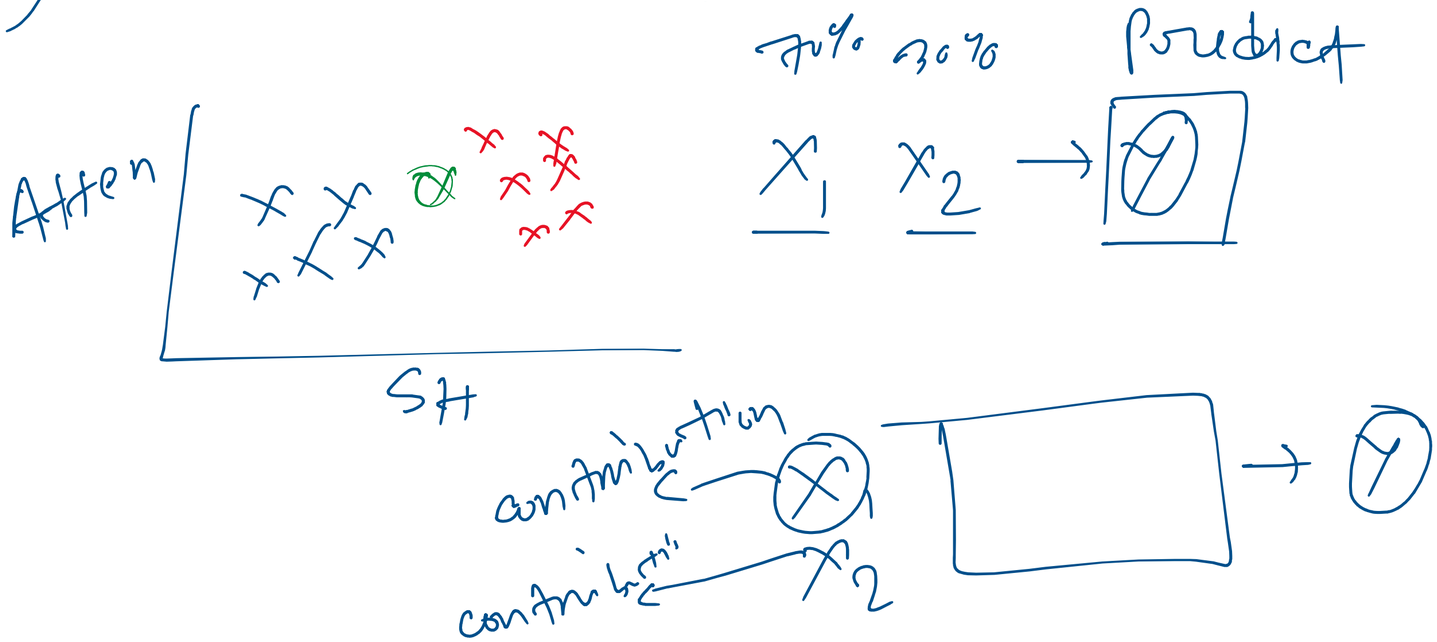
salary (0, 100000) →

5) Imbalance dataset

Yes — 95%

No — 5%

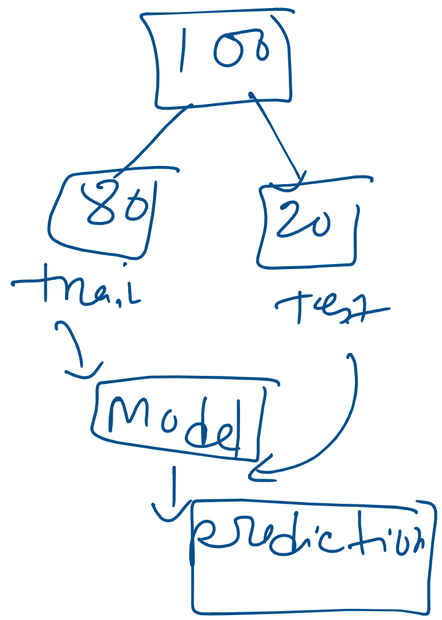
6) Black box Model



classification Report ✓

Accuracy

SH	Art	Actual Pass	Predict Pass
←	—	1	1 ✓
—	—	0	0 ✓
—	—	1	0 ✗
—	—	1	1 ✓
—	—	0	0 ✓



$$\text{Accuracy} = \frac{\text{number correct predict}}{\text{total number of Prediction}}$$

$$= \frac{4}{5}$$

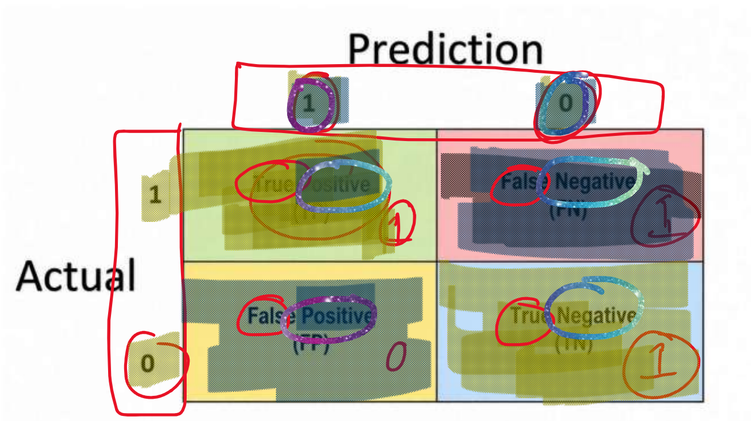
$$= \boxed{0.80} = \frac{80\% \rightarrow \text{correct}}{20\% \rightarrow \text{wrong}}$$

wrong

Actual Pass	Predict
error-1 → 0	1 ✗

✓ error-2 → 1 0 ✗

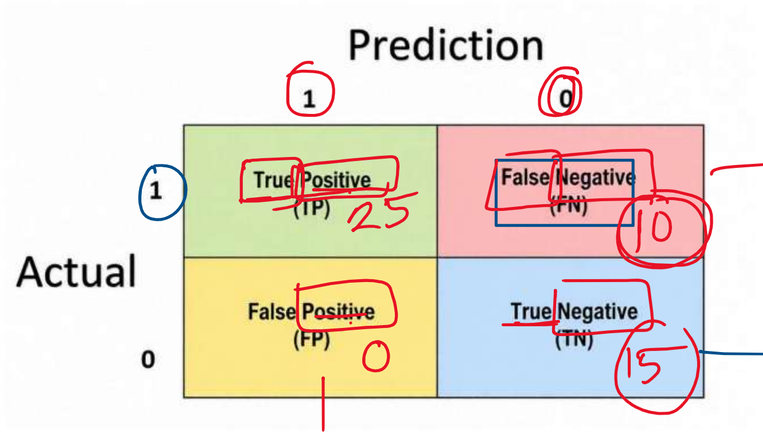
confusion Matrix



Act	Model Predic
1	0 ✗
0	0 ✓ ✓
1	1 ✓ ✓

TP + TN = 1 + 1 = 2

Heart disease
↳ Yes → 1
↳ No → 0

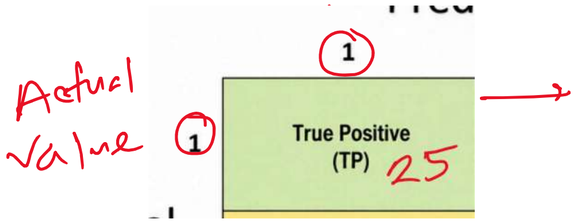


Model Pred → 0
Actual → 1

Model Pred → 0
Actual → 0 ✓

M
Prediction

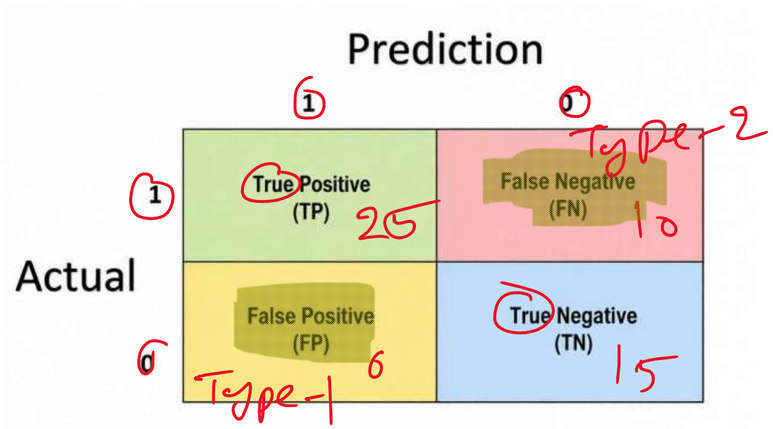
Prediction



Mode Predict → (1) Yes
Actual → (1) Yes

Target
0
1
2
3

	0	1	2	3
0				
1				
2				
3				



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

✓ confusion Matrix → Accuracy

✗ Accuracy → confusion Matrix

Precision

Prediction

Actual		Spam	Not Spam
	Spam	100	170
	Not Span	30	600

	Spam	Not Spam
Spam	100	190
Not Span	10	600

Model-A

Prediction

Model-B ✓

Actual		Spam	Not Spam
	Spam	100	FN 170
	Not Span	30	600

	Spam	Not Spam
Spam	100	FN 190
Not Span	10	600

ModelA = 700

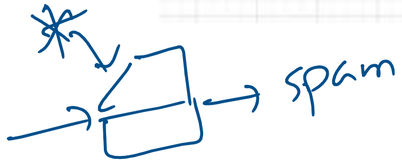
Actual			
	Spam	100	FN 170
	Not Span	FP 30	600

Spam	100	FN 190
Not Span	FP 10	600

$= \frac{1}{900}$

Model B

$= \frac{700}{900}$

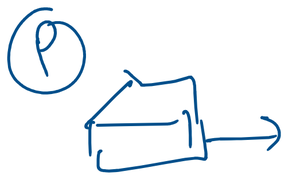


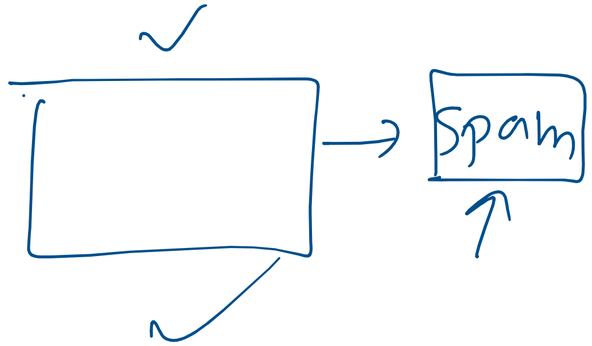
$FP_A > FP_B$

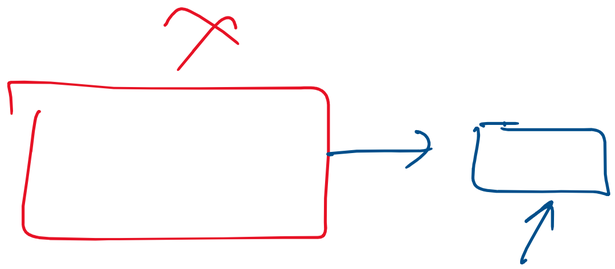
→ Fp = Model Predict → Spam
Actualy → Not Spam

→ FN = Model Predict → Not Spam
Actualy → Spam

Dangerous







Precision

Model-B

What Proportion of predicted Positives is truly positive

	Sent to Spam	Not sent to spam
Spam	True Positive	False Negative
Not Spam	False Positive	True Negative

Precision= TP/(TP+FP)

Prediction

Actual		Spam	Not Spam
	Spam	100	170
	Not Span	30	600

		Spam	Not Spam
	Spam	100	190
	Not Span	10	600

$$\text{Precision}_A = \frac{100}{100+30}$$

$$\text{Precision}_B = \frac{100}{100+10}$$

Recall

Prediction

Actual		Detected Cancer	Not Cancer
	Has Cancer	2000	200
	No Cancer	800	8000

		Detected Cancer	Not Cancer
	Has Cancer	2000	500
	No Cancer	500	8000

model-A ✓ Prediction model-B ✗

Actual		Detected Cancer	Not Cancer
	Has Cancer	2000	FN 200
	No Cancer	FP 800	8000

Actual		Detected Cancer	Not Cancer
	Has Cancer	2000	FN 500
	No Cancer	FP 500	8000

$FN_A < FN_B$

type-1 ⟶ FP → Model Predict → cancer ✓
~~Actual~~ → Not cancer

type-2 ⟶ FN → Model Predict → Not cancer ✗ ↓
Actual → cancer

Recall

What Proportion of Actual Positives is correctly classified?

$\frac{TP}{TP + FN}$

	Detected Cancer	Not detected Cancer
Has Cancer	True Positive	False Negative
No Cancer	False Positive	True Negative

Recall= TP/(TP+FN)

model-A ✓ Prediction model-B ✗

Actual		Detected Cancer	Not Cancer
	Has Cancer	2000	FN 200
	No Cancer	FP 800	8000

Actual		Detected Cancer	Not Cancer
	Has Cancer	2000	FN 500
	No Cancer	FP 500	8000

type 2

$Recall_A = \frac{2000}{2000 + 200}$ (200 = FN)

		Detected Cancer	Not Cancer
Actual	Has Cancer	2000	FN 200
	No Cancer	FP 800	8000

	Detected Cancer	Not Cancer
Has Cancer	2000	FN 500
No Cancer	FP 500	8000

$$\text{Recall}_B = \frac{2000}{2000 + 500}$$

FN

Model-A → Recall ✓

F_1 - score

model → Dog

model → Cat

Actual	Predicted
Dog	Cat
Cat	Dog

error → 1

error - 2

P & R → $\frac{P+R}{2}$

F_1 score → combination of Recall and Precision

$$F_1 \text{ score} = \frac{2PR}{}$$

$$F_1 \text{ score} = \frac{\cdots}{P+R}$$

F_1 score → Precision and Recall का

Harmonic Mean

$$\text{Mean} = \frac{P+R}{2}$$

$$F_1 \text{ score} = \frac{2*P*R}{P+R}$$

P=0 R=100

$$\text{mean} = \frac{0+100}{2} = 50$$

$$F_1 \text{ score} = \frac{2*0*100}{0+100} = 0$$

Model-A

P=70 R=70

Mean = 70

$$F_1 \text{ score} = \frac{2\times70\times70}{140} = 70$$

P=70 R=70

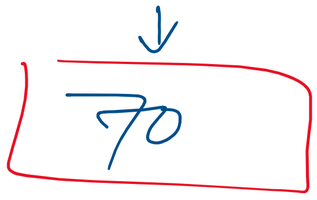
Model-B

P=40 R=100

Mean = 70

$$F_1 \text{ score} = \frac{2\times40\times100}{140} = \frac{4000}{70} = 57.14$$

P = 40 R = 10



Model A

= 57.14

P = 40 R = 100

